(Ridge Regression)

- Regularization
  - → Ridge
  - → Lasso.

$y = 0._ + 1.2x_1 + 20x_2 + 39x_3$

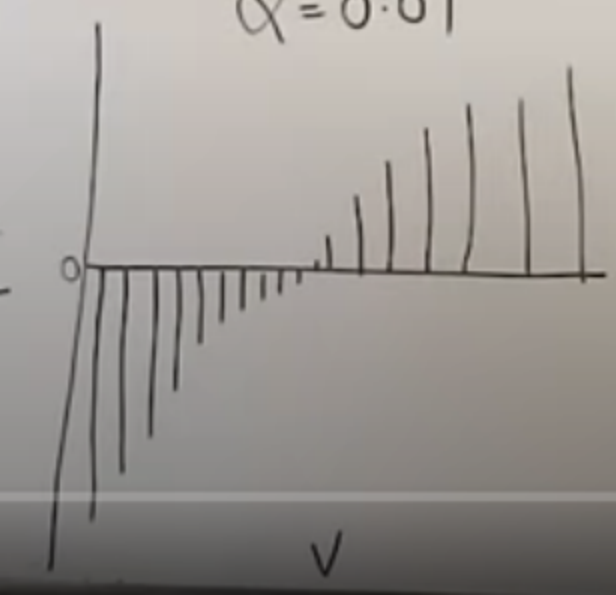
$y = 0.9 + 0.7x_1 + 2x_2 + 5x_3$

$\Rightarrow$ Ridge. $R = \text{Loss} + \alpha \|W\|^2$

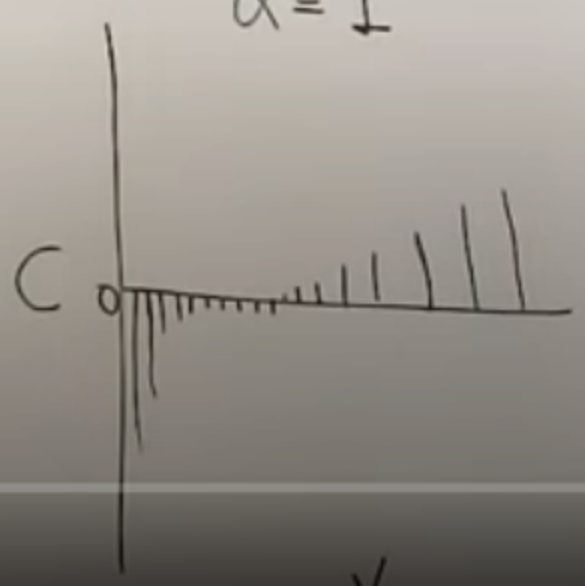
(penalty)

$\|W\|^2 = w_1^2 + w_2^2 + w_3^2 + \cdots + w_n^2$

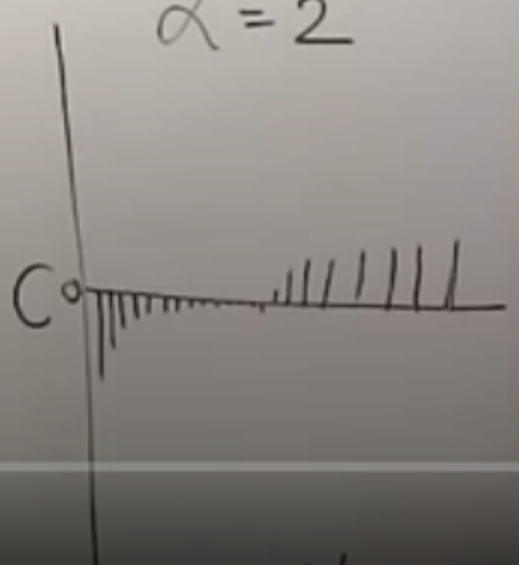
$\alpha = 0.01$



$\alpha = 1$



$\alpha = 2$



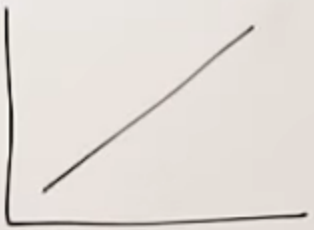
$$y = 0.7 + 1.2x_1 + 20x_2 + 39x_3$$

$$y = 0.9 + 0.7x_1 + 2x_2 + 5x_3$$

- Regularization
  - Ridge
  - Lasso.

(Ridge Regression)

- Regularization
  - → Ridge
  - → Lasso.



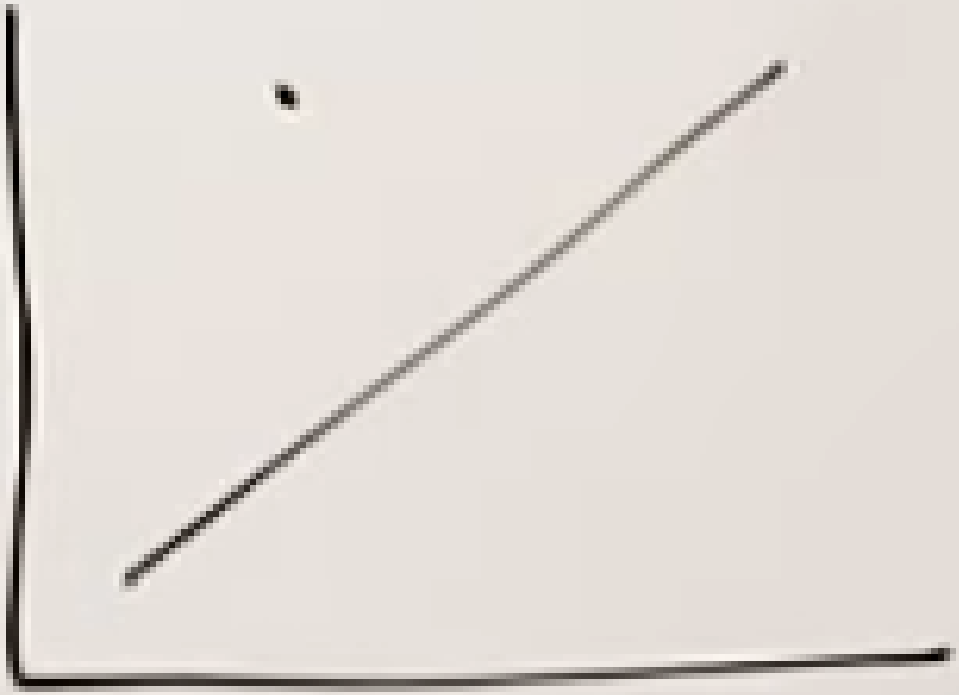
$y = 0.1 + 1.2x_1 + 20x_2 + 39x_3$

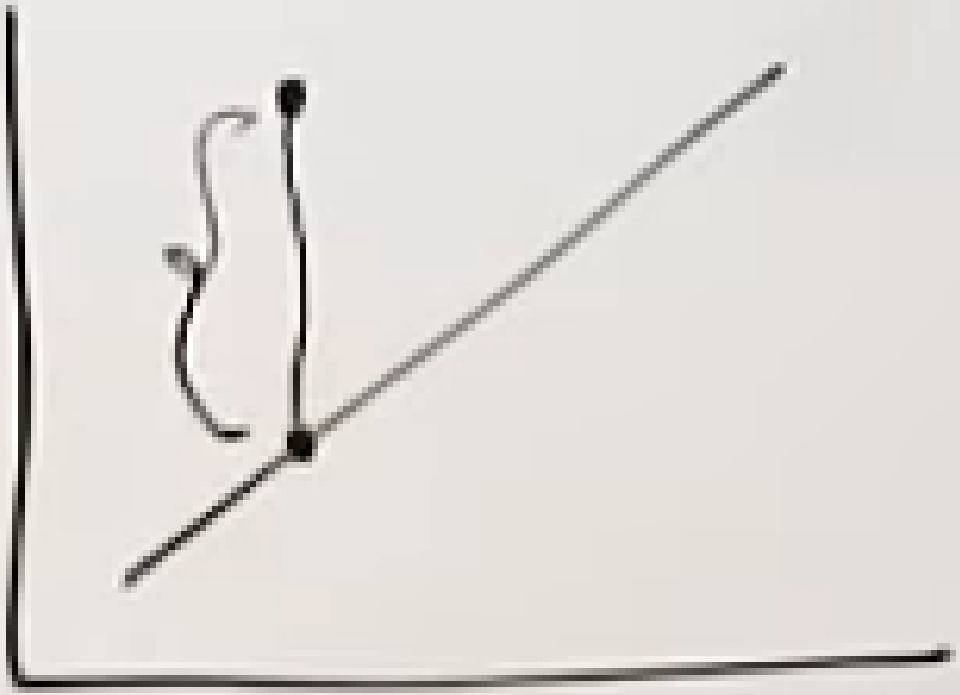
$y = 0.9 + 0.7x_1 + 2x_2 + 5x_3$

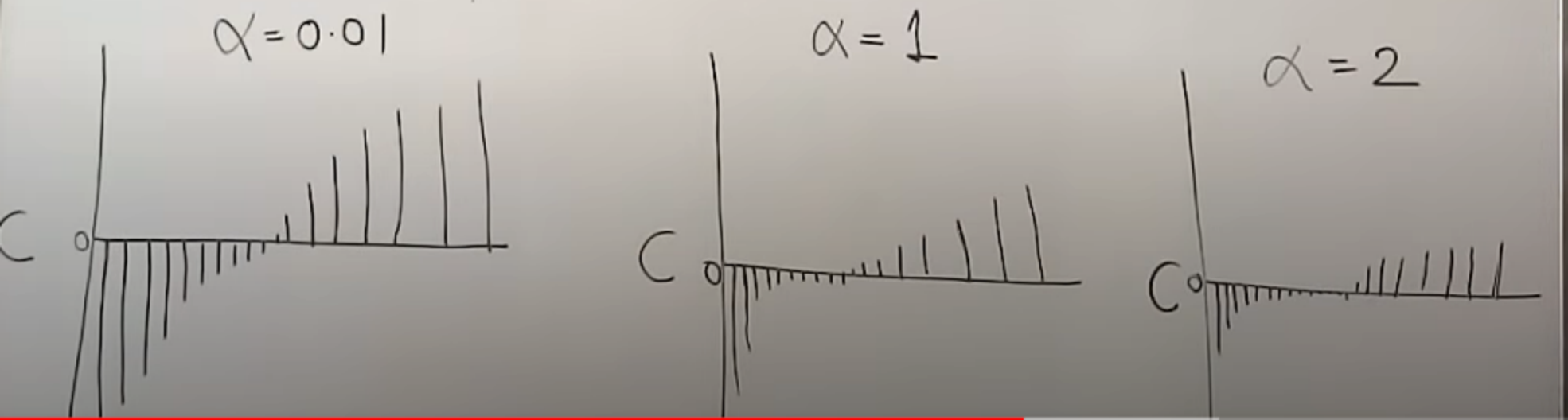
$\Rightarrow$ Ridge. R = Loss + $\alpha \|W\|^2$

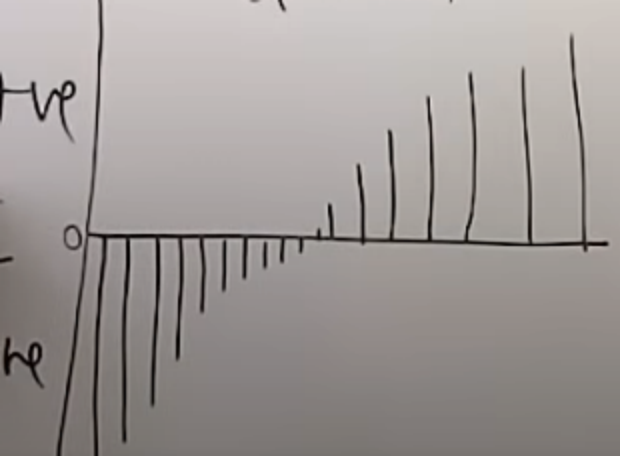
(penalty)

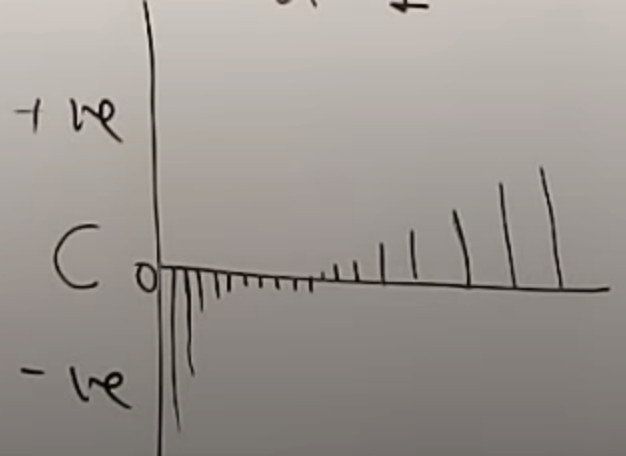
$$\|W\|^2 = w_1^2 + w_2^2 + w_3^2 + \cdots + w_n^2$$



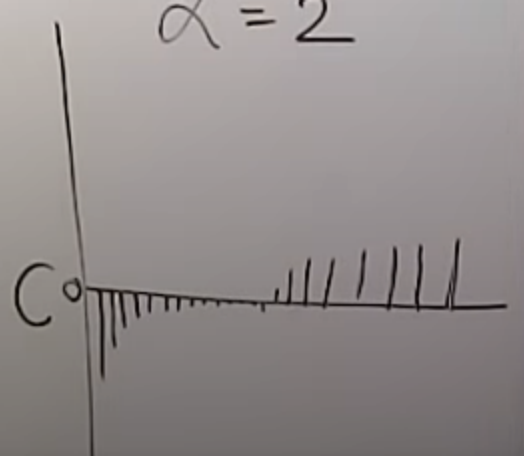



α = 0·01
C
0
α = 1
C
0
α = 2
C
0

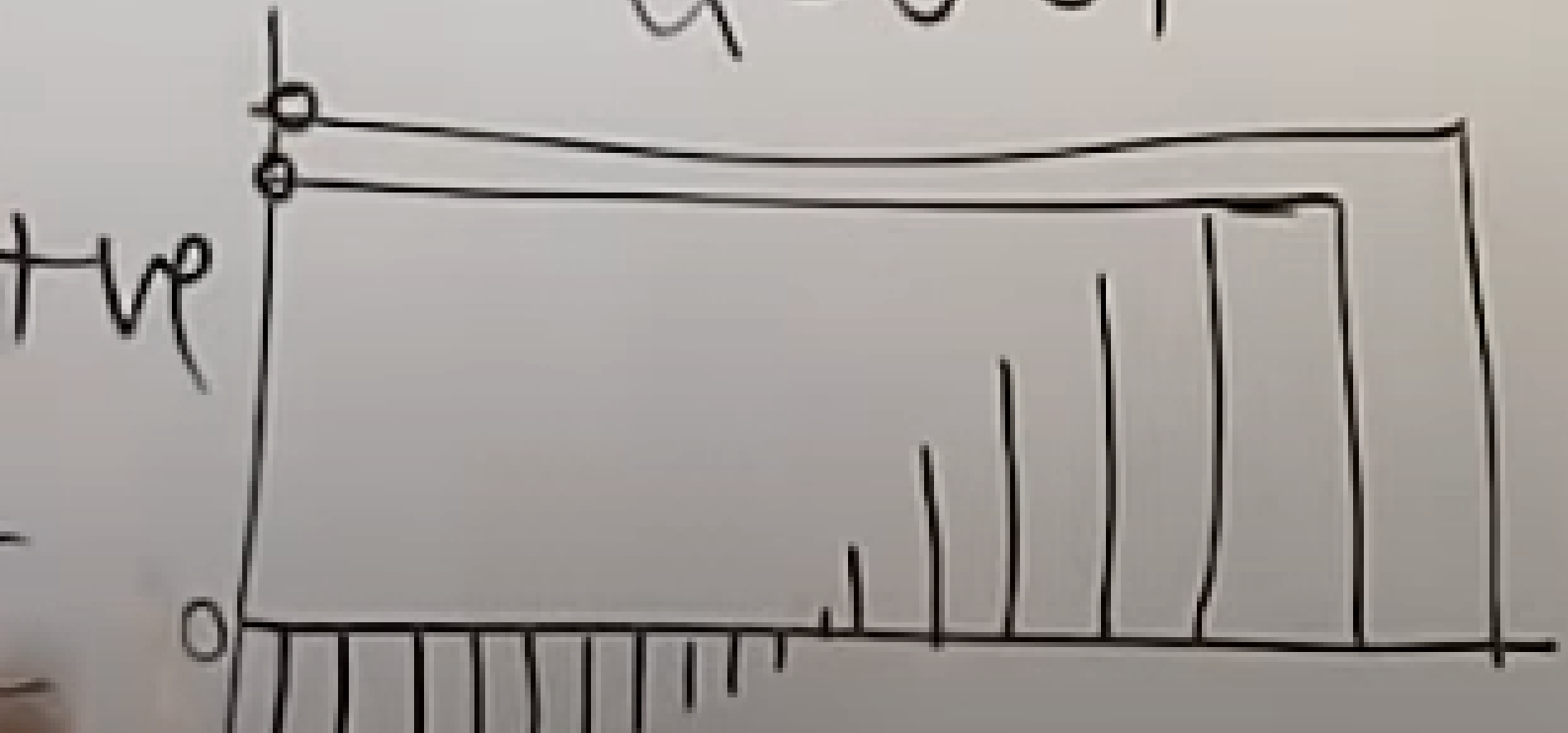




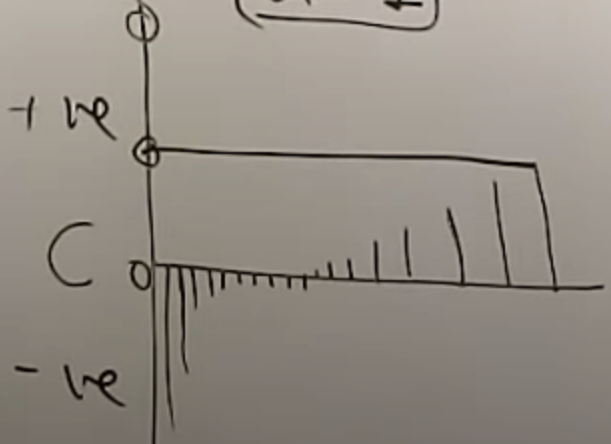
$\alpha = 2$



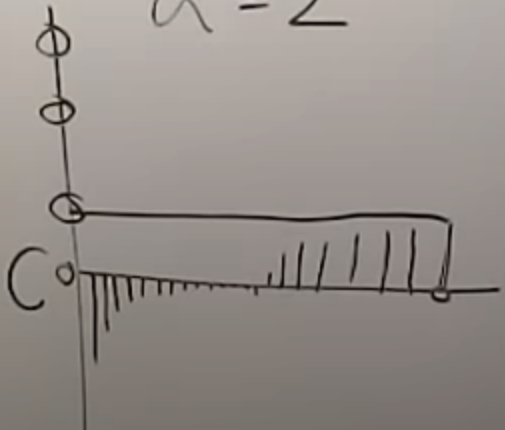
$\alpha = 0.01$

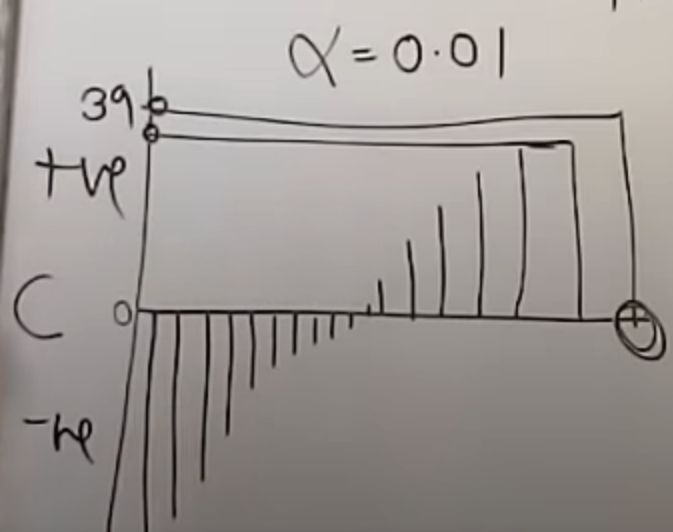


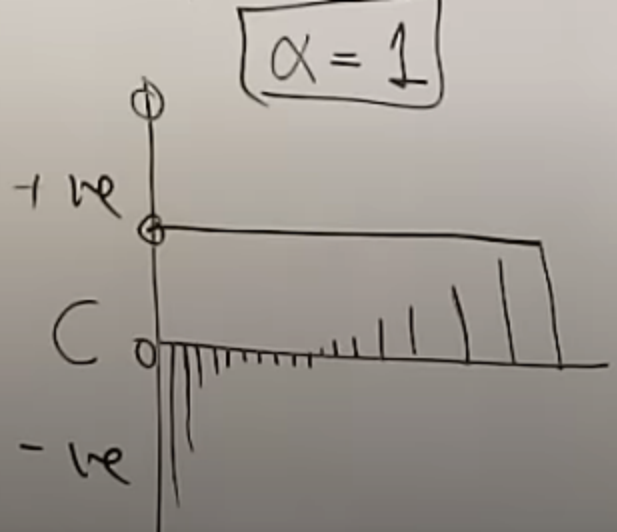
$\alpha = 1$

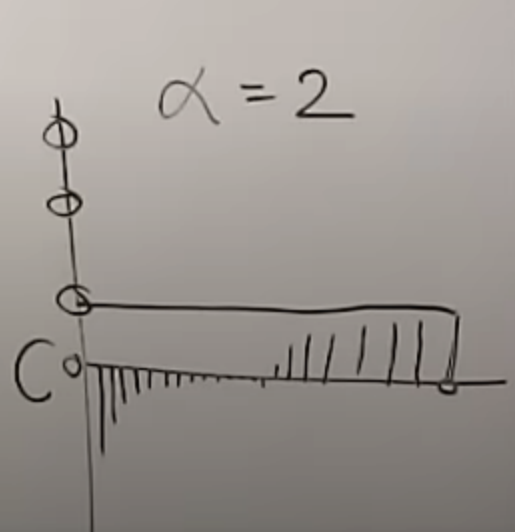


$\alpha = 2$

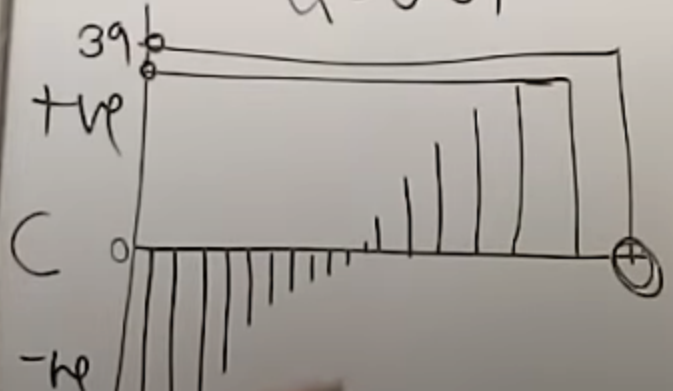



α = 0·01
39
+ve
C
-ve

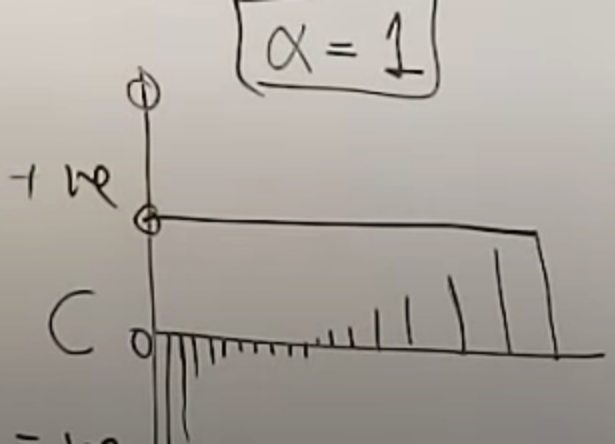

α = 1
+ve
C
-ve


α = 2
C

$\alpha = 0.01$



$\alpha = 1$



$\alpha = 2$

